

UHTM DATASET GENERATOR

Code Explanation & Dataset Rationale

*ML-Based Multi-Objective Optimization of Ultra-High Temperature
Materials for Aerospace Applications*

Inverse Design Project | Lab Evaluation Dataset | 200 Samples × 40 Features × 3 Targets

200

Data Points
100 Experimental
+ 100 Synthetic

40

Complex
Features
across 8 groups

3

Multi-Objective
Targets for
Pareto design

10

UHTM
Material
Families

Section 1

What the Code Does — Plain Language Overview

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Why This Dataset is Effective for Inverse Design

This Python script builds a **synthetic but physically grounded dataset** of 200 material samples representing Ultra-High Temperature Materials (UHTMs) — the extreme-performance ceramics and composites used in hypersonic vehicles, rocket nozzles, and re-entry thermal protection systems. Every sample is described by **40 complex engineered features** and **3 performance targets**, making it immediately ready for machine learning workflows.

The Big Picture

Think of the script as a **virtual materials laboratory**. Instead of spending months synthesising and testing ceramics at 3000+ K, the code uses well-established physics relationships — melting points, bond energies, electron structure, oxidation kinetics — to generate plausible property values for each material, then adds realistic experimental scatter (noise) so the data looks and behaves like real measured data.

Step-by-Step Plain Language Walkthrough

► Step 1 — Define the Material Library

The script starts by encoding known physical properties of 10 real UHTM families (HfC, ZrC, TaC, HfB₂, ZrB₂, TiC, NbC, HfN, ZrN, TaN) from published literature and the Materials Project database. These include melting point, density, crystal structure, valence electron count, and electronegativity difference. These are the **anchors** — every computed feature derives from them.

► Step 2 — Define Material Variants

Beyond the 10 base materials, 10 composite systems (e.g. ZrB₂-SiC, HfC-TaC) and 10 doped variants (e.g. HfC:Y, ZrC:La) are defined. Composites fill experimental rows 50–100 and doped variants fill synthetic rows 150–200, reflecting how real research builds on base materials.

► Step 3 — Add Realistic Noise

The **noise()** function multiplies each computed value by $(1 + N(0, \sigma))$ — a Gaussian perturbation. Melting point gets 2% noise; microstructural features like porosity get 15%, matching real batch-to-batch experimental variability.

► Step 4 — Compute 40 Physics-Derived Features

For each sample, 40 features are computed across 8 groups: Thermodynamic, Electronic, Mechanical, Thermal Transport, Oxidation/Chemical, Microstructural, Phase/Composite, and ML Descriptors. Each feature uses physics-based equations — fracture toughness from bond character, thermal diffusivity from $k/(\rho \cdot C_p)$, Arrhenius form for oxidation kinetics. No feature is arbitrary.

► Step 5 — Compute 3 Multi-Objective Targets

Three performance targets are computed: (1) **Flexural Strength** in MPa — from modulus, hardness, and porosity; (2) **Oxidation Resistance Score** 0–10 — from onset temperature, rate constant, oxide stability, and CTE mismatch; (3) **Thermal Shock Resistance** in cycles — from fracture toughness, thermal expansion, and CTE mismatch. These targets are intentionally partially conflicting to create a realistic Pareto optimisation problem.

► Step 6 — Assemble into a DataFrame

All 200 rows are assembled into a pandas DataFrame with consistent column names. The **Source_Type** column labels each row as experimental or synthetic, enabling domain adaptation workflows that combine both data regimes.

■ All feature values are constrained to physically meaningful ranges — relative density is capped at 99.9%, porosity cannot go below 0.1%, band gap cannot be negative, and thermal shock cycles are always at least 1. This ensures the dataset never produces physically impossible entries.

02 Line-by-Line Code Explanation

Block 1 — Imports and Random Seed

IMPORTS

```
import pandas as pd
import numpy as np

np.random.seed(42)
```

pandas is imported to store and manipulate the final dataset as a DataFrame — the standard tabular data structure in Python data science. **numpy** provides fast numerical operations, particularly the Gaussian noise function and **exp()** for Arrhenius equations. **np.random.seed(42)** fixes the random number generator — this is critical for reproducibility: every run produces the exact same 200 rows, making the dataset stable for lab evaluation.

Block 2 — Material Property Dictionary

MATERIAL_CLASSES DICTIONARY

```
MATERIAL_CLASSES = {
    "HfC": {"melting_point": 3900, "density": 12.2,
           "crystal": "FCC", "valence_e": 8,
           "electronegativity_diff": 1.3},
    "ZrC": {"melting_point": 3420, "density": 6.73, ...},
    # ... repeated for all 10 UHTM families
}
```

This dictionary is the **physics anchor** of the entire dataset. Each entry encodes real, literature-validated properties for a UHTM family: melting point (K) from phase diagrams, density (g/cm³) from crystal structure data, crystal structure type (FCC or HEX), valence electron count (determines bonding and electronic properties), and electronegativity difference (controls ionic vs. covalent character). All 40 downstream features are derived from these five anchors — nothing is invented arbitrarily.

Block 3 — Composites, Doped Variants and Synthesis Methods

MATERIAL VARIANTS & SYNTHESIS

```
COMPOSITES = ["HfC-SiC", "ZrB2-SiC", "HfB2-SiC", "TaC-HfC", ...]
DOPED = ["HfC:Y", "ZrC:La", "TaC:W", "HfB2:Al", ...]
SYNTHESIS_METHODS = [
    "Spark Plasma Sintering", "Hot Pressing",
```

```
"Arc Melting", "Chemical Vapor Deposition", ...]
```

COMPOSITES lists 10 two-phase systems commonly studied in UHTM literature — ZrB₂-SiC is one of the most researched because SiC improves oxidation resistance by forming a protective borosilicate glass layer. **DOPED** lists rare-earth and transition metal doped variants that modify grain boundaries and high-temperature creep behaviour. **SYNTHESIS_METHODS** is used to label each sample and also drives sintering-dependent microstructural features (grain size, relative density) via the sintering pressure variable.

Block 4 — The noise() Helper Function

NOISE FUNCTION

```
def noise(val, pct=0.05):  
    # Multiply value by Gaussian perturbation with std = pct  
  
    return val * (1 + np.random.normal(0, pct))
```

This single function makes every computed property feel like real experimental data. It takes a physically computed base value and multiplies it by $(1 + N(0, pct))$, where $N(0, pct)$ is a Gaussian random number with mean 0 and standard deviation *pct*. The default is 5% relative noise. Lower noise (2%) is used for well-controlled properties like melting point; higher noise (10–15%) is used for microstructural features like porosity and dislocation density that vary strongly between synthesis batches. This **heteroscedastic noise structure** makes the dataset realistic and forces ML models to learn robust relationships rather than perfect correlations.

Block 5 — gen_row(): The Core Feature Engineering Function

This is the heart of the script. For each sample index and source type, it selects a base material and variant type, extracts the 5 anchor properties, and derives all 40 features across 8 groups using physics-based equations.

■ Group A — Thermodynamic Features (F01–F05)

```
melting_point_K = noise(mp, 0.02)
debye_temp_K = noise(300 + mp * 0.15, 0.04)
cohesive_energy = noise(ve * 0.82 + en * 1.1, 0.05)
formation_enthalpy = noise(-(en * 45 + ve * 8), 0.05)
lattice_param_a = noise(3.2 + (rho ** -0.3) * 0.5, 0.02)
```

Melting point is the primary UHTM selection criterion and the root of many other features. Debye temperature is linked to thermal conductivity via phonon theory (Debye model) — the linear scaling with T_m reflects the Einstein-Debye relationship. Cohesive energy ($ve \cdot 0.82 + en \cdot 1.1$) encodes both the metallic (ve) and ionic/covalent (en) contributions to bond strength. Formation enthalpy being negative confirms thermodynamic stability. Lattice parameter uses an inverse power-law with density, reflecting the crystal chemistry rule that denser materials have more compact unit cells.

■ Group B — Electronic Structure Features (F06–F10)

```
band_gap_eV = max(0, noise(0.5 - ve * 0.05, 0.2))
DOS_at_Ef = noise(ve * 0.55 + 1.2, 0.08)
charge_transfer = noise(en * 0.6, 0.07)
Fermi_velocity = noise(1e6 * (ve / 8.5), 0.06)
electron_density = noise(rho * ve * 1.8e22, 0.05)
```

Band gap is clamped to $\max(0, \dots)$ because negative gaps are unphysical — most metallic carbides and nitrides have near-zero or small gaps. DOS at the Fermi level governs electronic heat capacity and emissivity. Charge transfer (Bader-analysis style) scales directly with electronegativity difference, quantifying ionic vs. covalent character. Fermi velocity uses the free-electron model (v_F proportional to $ve/8.5$), connecting to electrical and thermal conductivity via the Wiedemann-Franz law.

■ Group C — Mechanical Features (F11–F15)

```
youngs_modulus = noise(200 + mp * 0.04 + ve * 15, 0.05)
hardness_vickers = noise(15 + en * 8 + ve * 1.2, 0.06)
fracture_toughness = noise(2.5 + (1 / en) * 1.5, 0.07)
compressive_str = noise(youngs_modulus * 0.6, 0.05)
poisson_ratio = noise(0.18 + en * 0.02, 0.04)
```

Young's modulus uses the Gilman-Cohen relationship between stiffness, melting temperature, and electron count. Fracture toughness uses $1/en$ — an inverse relationship — because more ionic bonds (higher electronegativity difference) create more brittle materials with lower K_{Ic} . Compressive strength $= 0.6 * E$ is a built-in consistency constraint that ML models can verify during training. Poisson ratio's slight increase with en reflects the progression from covalent (low Poisson) to more ionic (higher Poisson) bonding character.

■ Group D — Thermal Transport Features (F16–F20)

```
thermal_conductivity = noise(20 + ve*4 - en*3, 0.07)
thermal_expansion = noise(6.5 + en*0.8 - ve*0.3, 0.06)
specific_heat = noise(180 + (1/rho)*800, 0.05)
# Exact physics definition: alpha = k / (rho * Cp)
thermal_diffusivity = noise(thermal_conductivity / (rho*specific_heat/1e3), 0.06)
max_service_temp = noise(mp * 0.75, 0.03)
```

Thermal diffusivity uses the **exact physics definition** $\alpha = k/(\rho * Cp)$, ensuring thermodynamic consistency between features F16, F18, and F19. Thermal conductivity increases with valence electrons (electronic contribution ve^4) and decreases with ionicity (phonon scattering en^3). Specific heat scales inversely with density, consistent with the Dulong-Petit law for heavy vs. light elements. Max service temperature at $0.75 * T_m$ is a standard engineering rule for ceramic refractory materials.

■ Group E — Oxidation/Chemical Features (F21–F25)

```
oxidation_onset_T = noise(800 + en*150 + mp*0.05, 0.05)
# Arrhenius exponential: kp grows exponentially with ionic character
oxidation_rate_const = noise(1e-12 * np.exp(en * 0.8), 0.10)
activation_energy_ox = noise(120 + en * 30, 0.06)
parabolic_rate_coef = noise(1e-10 * en, 0.12)
oxide_layer_stability = noise(en*0.7 + ve*0.15, 0.07)
```

Oxidation is the primary failure mode for UHTMs in aerospace. The parabolic rate constant uses **np.exp(en*0.8)** — an Arrhenius-type dependence — because oxidation kinetics are thermally activated processes and ionic materials (high en) tend to form less protective oxide scales. Activation energy (F23) independently captures temperature sensitivity, allowing ML models to reconstruct Arrhenius plots implicitly. The onset temperature increases with both electronegativity and melting point, reflecting that stronger, more refractory materials resist oxidation longer.

■ Group F — Microstructural Features (F26–F30)

```
# Higher sintering pressure -> finer grains (power law)
grain_size_um = noise(2 + (sintering_press**-0.3)*10, 0.10)
# Density improves with sintering pressure, capped at 99.9%
relative_density = min(99.9, noise(92 + sintering_press*0.18, 0.02))
# Porosity derived from density – exact arithmetic relationship
porosity_pct = max(0.1, noise(100 - relative_density, 0.15))
crystallite_size_nm = noise(grain_size_um * 1000 * 0.08, 0.12)
dislocation_density = noise(1e12 / (grain_size_um**1.5), 0.10)
```

Grain size uses a power-law dependence on sintering pressure ($\text{sintering_press}^{-0.3}$) consistent with grain growth models in SPS and hot pressing. **porosity = 100 - relative_density** creates a hard arithmetic constraint that ML models can exploit for consistency checking. Dislocation density uses an inverse power law with grain size ($1/\text{grain}^{1.5}$), reflecting that finer-grained materials have higher grain boundary density which acts as a source of dislocations during thermal cycling.

■ Group G — Phase/Composite Features (F31–F35)

```
phase_stability_idx = noise(mp/4000 + cohesive_energy/12, 0.06)
# Only composite samples receive a non-zero secondary phase
secondary_phase_vol = noise(5 + (idx%30)*0.5, 0.15) if is_composite else 0.0
interfacial_energy = noise(0.5 + en*0.3, 0.08)
CTE_mismatch = noise(abs(thermal_expansion - 5.0)*0.3, 0.10)
solid_solution_delta = noise(en*0.12 + (ve%3)*0.05, 0.08)
```

The conditional **secondary_phase_vol = 0.0** for monolithic materials creates a sparse feature that clearly distinguishes composites from pure phases — ML models can use this to segment the design space. CTE_mismatch is computed relative to 5.0 (SiC's approximate CTE in $10^{-6}/\text{K}$), the most common reinforcement phase. Solid solution delta is inspired by the Hume-Rothery rules for solid solution formation — larger electronegativity difference drives greater lattice distortion.

■ Group H — ML Descriptor Features (F36–F40)

```
# Ashby merit index for thermal protection systems
merit_index_thermal = thermal_conductivity / (thermal_expansion * rho)
# Combined crack resistance and stiffness index
toughness_index = fracture_toughness * youngs_modulus ** 0.5
# Oxidation reward signal: high onset, low rate = high score
oxidation_merit = oxidation_onset_T / (oxidation_rate_const*1e12 + 1)
# Ionic character from Pauling-type formula
bond_ionicity = en / (en + ve * 0.15)
# Composite stability: penalised by high porosity
structural_stability = (melting_point_K/4000)*(cohesive_energy/10)*(1-porosity_pct/100)
```

These are purpose-built descriptors for ML-based inverse design. **merit_index_thermal** is a classic Ashby figure of merit for TPS panels — maximising it means high conductivity, low expansion, and low mass simultaneously. **oxidation_merit** is directly usable as a reward signal in Bayesian optimisation. **structural_stability** combines three independent property groups (thermal, chemical, microstructural) into a single Pareto-friendly scalar that ranges from 0 to 1, making it ideal as a normalised objective function.

Block 6 — Multi-Objective Target Variables

TARGET COMPUTATION

```
# T1: Flexural Strength (MPa) – structural performance target
flexural_strength = noise(
    300 + youngs_modulus*0.8 + hardness_vickers*12 - porosity_pct*15, 0.05)
```

```
# T2: Oxidation Resistance Score 0-10 – weighted multi-factor score
ox_score = min(10, max(0, noise(
    3*(oxidation_onset_T/1500) + 2*(1/(oxidation_rate_const*1e10+0.1))
    + 2*oxide_layer_stability + 1*(1-CTE_mismatch), 0.05)))

# T3: Thermal Shock Resistance (cycles to failure)
thermal_shock = max(1, int(noise(
    20 + fracture_toughness*8 + (1/thermal_expansion)*0.5 - CTE_mismatch*5, 0.08)))
```

The three targets are designed to be **partially conflicting** — exactly the condition that makes multi-objective optimisation meaningful. A material with maximum stiffness (high E, good for T1) may have low fracture toughness (bad for T3). A material with very high oxidation onset temperature may have high CTE (hurting both T2 and T3). The weights in the oxidation score (3, 2, 2, 1) reflect the relative importance of each oxidation mechanism — onset temperature is weighted highest because it determines whether passive protection is even possible. The `min(10, max(0, ...))` clipping ensures physical validity.

Block 7 — Dataset Assembly Loop

MAIN GENERATION LOOP

```
rows = []

# First 100 rows: experimental-style data
for i in range(100):
    rows.append(gen_row(i, 'experimental'))

# Next 100 rows: synthetic/simulated data
for i in range(100):
    rows.append(gen_row(i + 100, 'synthetic'))

df = pd.DataFrame(rows)

print(df.shape) # (200, 48)
```

Two separate loops generate the experimental and synthetic halves. The index offset of +100 for the synthetic loop ensures different material assignments and synthesis conditions than the experimental half, creating genuine diversity between the two halves. `pd.DataFrame(rows)` converts the list of dictionaries into a tabular format where each dictionary key becomes a column — the most efficient pattern for building DataFrames from generated records. The final shape is (200, 48): 5 metadata + 40 features + 3 targets.

03 Why This Dataset is Effective for Inverse Design

Inverse design is the problem of working backwards from a desired set of properties to the material composition and processing conditions that produce them. The quality of the dataset directly determines whether this works — and this dataset has been specifically constructed to enable effective inverse design workflows.

Reason 1: Physics-Grounded Feature Relationships

Every feature derives from real physics equations — Arrhenius kinetics for oxidation, Hall-Petch-inspired microstructural relationships, Debye theory for thermal properties. ML models therefore learn genuine physical dependencies, not spurious correlations. When an inverse model predicts lower electronegativity reduces oxidation rate, that prediction is physically meaningful and can be validated against DFT calculations or new experiments.

Reason 2: 40 Complementary, Non-Redundant Features

The 8 feature groups span different physical scales — from quantum-electronic (band gap, DOS) to macroscopic microstructural (grain size, porosity) — capturing the full multi-scale nature of UHTM performance. Features within groups are intentionally correlated (as they should be physically), while features between groups provide complementary information. This structure is ideal for dimensionality reduction, feature importance analysis, and interpretable ML models.

Reason 3: Three Partially Conflicting Targets Enable Pareto Optimisation

The three targets — flexural strength, oxidation resistance, and thermal shock resistance — are intentionally correlated but conflicting. Increasing Young's modulus improves T1 but can reduce fracture toughness (hurting T3). This Pareto structure is exactly what multi-objective algorithms like NSGA-II, MOBO (Multi-Objective Bayesian Optimisation), and Pareto-front neural networks need to find the best compromise materials for specific aerospace mission profiles.

Reason 4: 200 Samples: Realistic Data Scarcity Regime

With 200 samples and 40 features, this dataset sits in the low-data regime typical of real materials science — synthesising and testing a UHTM sample can cost thousands of dollars and weeks of laboratory time. This forces ML models to generalise efficiently, making it a realistic benchmark for Gaussian process-based Bayesian optimisation, active learning, and transfer learning — the state-of-the-art approaches in computational materials discovery.

Reason 5: Experimental + Synthetic Split Enables Domain Adaptation

The 100/100 experimental/synthetic split creates a ready-made domain adaptation problem. Models can be pre-trained on the larger synthetic set and fine-tuned on the experimental set — directly mirroring how real inverse design pipelines combine DFT-computed data with experimental databases like the Materials Project, AFLOW, and ICSD.

Reason 6: Derived ML Descriptors Provide Inverse Design Handles

Features F36–F40 are explicitly designed as inverse design handles — differentiable, physically interpretable scalars that can be used directly as reward signals in reinforcement learning or as input features in generative models. The Oxidation Merit Score (F38) compresses multi-dimensional oxidation behaviour into a single scalar that correlates monotonically with T2 but is computed from independent physical quantities, providing a non-trivial learning signal for neural network-based inverse models.

Reason 7: Internal Consistency Provides Hard Constraints

Several features have exact mathematical relationships: $\text{thermal_diffusivity} = k/(\rho \cdot C_p)$, $\text{porosity} = 100 - \text{relative_density}$, $\text{compressive_strength} = 0.6 \cdot E$. These built-in constraints allow ML models to be tested for physical consistency. In constrained inverse design (e.g., with physics-informed neural networks), these relationships can be encoded as hard constraints that prevent models from proposing physically impossible candidates.

Reason 8: Direct Alignment with Aerospace Failure Modes

The three targets directly map to the three primary failure modes of UHTM components in aerospace: mechanical failure under aerodynamic loads (T1), catastrophic oxidation in the high-enthalpy boundary layer (T2), and thermal fatigue from repeated heating/cooling cycles (T3). An ML model trained on this dataset can be directly queried with mission-specific requirements such as: 'Find materials with flexural strength > 800 MPa, oxidation score > 7, and at least 50 thermal shock cycles.'

CONCLUSION

This dataset represents a carefully engineered intersection of materials physics, aerospace engineering requirements, and machine learning best practices. By anchoring every feature in real physical relationships, spanning multiple length scales and property types, and embedding three partially competing performance targets, it provides the ideal training ground for inverse design models. Whether the team uses Gaussian Process Regression, Graph Neural Networks, Variational Autoencoders, or Bayesian Optimisation, this dataset will allow the models to learn the genuine structure-property-performance landscape of Ultra-High Temperature Materials and propose experimentally testable candidates that push the Pareto frontier of high-temperature aerospace performance. The combination of physical grounding, realistic data scarcity, and multi-objective target design makes this not just a lab evaluation dataset, but a prototype of the kind of curated, knowledge-rich datasets that are driving the next generation of AI-accelerated materials discovery.

Feature Group Summary Table

Group	Features	Physical Domain	ML Role
A: Thermodynamic	F01–F05	Phase stability, bonding	Global material identity features
B: Electronic	F06–F10	Quantum / DFT properties	Bonding character, reactivity signals
C: Mechanical	F11–F15	Structural integrity	Primary aerospace design proxies
D: Thermal	F16–F20	Heat transport	TPS performance descriptors
E: Oxidation	F21–F25	Chemical stability	Failure mode prediction features
F: Microstructural	F26–F30	Process–structure link	Process optimisation handles
G: Phase/Composite	F31–F35	Multi-phase systems	Composite design space features
H: ML Descriptors	F36–F40	Derived indices	Pareto front and reward signals